Section 1: NASA C-MAPSS Simulation Architecture and Operating Context

The Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) is a high-fidelity turbofan engine degradation modeling tool developed by NASA Glenn Research Center.

The simulation model reflects a 90,000-lb thrust class twin-spool turbofan engine operating across varied operational regimes including sea level static, climb, cruise, and descent.

1.1 Engine Spool Topology:
- Low Pressure Spool: Fan (driven by Low Pressure Turbine, LPT) and Low Pressure Compressor (LPC). Monitored by physical fan speed sensor s_8 (Nf, nominal 2388 rpm) and corrected fan speed s_13 (NRf).
- High Pressure Spool: High Pressure Compressor (HPC) driven by High Pressure Turbine (HPT). Monitored by core speed sensor s_9 (Nc, nominal 9065 rpm) and corrected core speed s_14 (NRc, nominal 8143 rpm).

1.2 Operating Regimes and Environmental Envelope:
Subsets FD001 and FD003 represent engines operating under a single nominal flight condition (sea level static). Subsets FD002 and FD004 encompass complex multi-regime profiles featuring 6 distinct flight operating clusters defined by altitude (0 to 42,000 ft), Mach number (0.0 to 0.84), and throttle resolver angle (TRA 60 to 100 degrees). Multi-regime engines require regime standardization before degradation assessment.

Section 2: Comprehensive Sensor Mapping and Baseline Operating Ranges

The C-MAPSS telemetry acquisition stream records 21 continuous sensor parameters plus 3 operational setting channels:

- s_1 (T2): Total temperature at fan inlet. Nominal: 518.67 R. Flat-line sensor in single-regime datasets.
- s_2 (T24): Total temperature at LPC outlet. Healthy baseline: 642.68 R. Warning threshold: > 643.50 R (indicates LPC efficiency loss or blade fouling).
- s_3 (T30): Total temperature at HPC outlet. Healthy baseline: 1590.52 R. Critical threshold: > 1605.0 R (strong indicator of HPC blade degradation and hot-section distress).
- s_4 (T50): Total temperature at LPT outlet (Exhaust Gas Temperature - EGT). Healthy baseline: 1408.93 R. Critical threshold: > 1425.0 R. EGT margin deterioration directly correlates with remaining useful life.
- s_6 (P15): Total pressure in bypass duct. Nominal: 21.61 psia.
- s_7 (P30): Total pressure at HPC outlet. Healthy baseline: 553.37 psia. Decreases as compressor blade tips erode and clearances open up.
- s_8 (Nf): Fan physical speed. Nominal: 2388.10 rpm.
- s_9 (Nc): Core physical speed. Healthy baseline: 9065.24 rpm. Increases abnormally when core components degrade to maintain required thrust output.
- s_11 (Ps30): Static pressure at HPC outlet. Baseline: 47.54 psia. Downward drift (> 0.5 psia drop) indicates HPC flow capacity degradation.
- s_12 (phi): Fuel flow to Ps30 ratio. Baseline: 521.66 pps/psi.
- s_15 (BPR): Bypass Ratio. Baseline: 8.44. Drifts upward as high-pressure core swallows less air due to constriction.
- s_17 (htBleed): High-pressure bleed enthalpy. Baseline: 393.20 Btu/lbm.
- s_20 (W31): HPT coolant bleed flow. Baseline: 38.81 lbm/s.
- s_21 (W32): LPT coolant bleed flow. Baseline: 23.29 lbm/s.

Section 3: Failure Mechanisms and RUL Piecewise Progression

3.1 High Pressure Compressor (HPC) Degradation:
The predominant failure mode in C-MAPSS FD001 and FD002 is progressive HPC fouling and tip rubbing. Degradation manifests as simultaneous loss in flow capacity (up to -5%) and isentropic efficiency (up to -3%).

3.2 Fan and Low Pressure Section Degradation:
In FD003 and FD004, a secondary failure mode involves Fan degradation and LPC clearance increase, characterised by early divergence in s_2 (T24) and s_8 (Nf) rotor flutter.

3.3 Piecewise Linear Target RUL Modeling:
In early operational cycles, health degradation is undetectable. The standard NASA prognostic benchmark caps the ground truth Remaining Useful Life (RUL) at 125 cycles. Beyond the inflection cycle, RUL decays linearly to 0 at failure.
Predicted RUL Risk Categories:
- RUL > 85 cycles: Low Risk (Health Score 80-100%). Routine telemetry tracking.
- RUL 50-85 cycles: Medium Risk (Health Score 60-79%). Plan inspection in next scheduled window.
- RUL 25-50 cycles: High Risk (Health Score 35-59%). High priority inspection; order replacement parts.
- RUL < 25 cycles: Critical Risk (Health Score 0-34%). Immediate operational safety alert and dispatch.